National University of Computer and Emerging Sciences
Peshawar Campus

Artificial Intelligence (AI2002) / App. Artificial Intelligence (AI4007)

Sessional-I Exam

Date: 22nd February 2025

Course Instructor(s)

Shahzeb Khan / Waqas Ali

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 3

____________ Roll No   ____________ Section   __________________ Student Signature

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Attempt all the questions.

CLO 1: Summarize the notions of rational behavior and intelligent agents.

Q1: A company is developing an autonomous delivery drone system. The drones need to navigate through a city, avoiding obstacles like buildings and trees while delivering packages. Weather conditions such as wind and rain can affect their flight path. The system must decide the best route while ensuring timely and safe deliveries. **[10]**

a) **What type of environment does this problem represent?** Explain whether it is deterministic or stochastic, and episodic or sequential, with reasons. **[4]**

b) **What type of agent is best suited for this task?** Should it be a simple reflex agent, reflex agent with state, goal-based agent, or utility-based agent? Justify your answer. **[3]**

c) **What performance measures should be used to evaluate the drone's effectiveness?** Provide at least three relevant performance criteria and explain their importance. **[3]**

CLO 3: Apply methods of blind as well as informed search on the problem related to the programs.

Q2: Recently, the FAST Sports Society organized a sports week featuring multiple events. Among them, the **Tug of War** gained the most attention. However, with several games happening simultaneously, students faced difficulty navigating to the event venue. **[10]**

The Head of the Sports Society asks for your help to find the best path(s) from **Gate (A) to Event Place (G) (**Graph shown in Figure 1). Since multiple routes may exist, ties should be broken in **alphabetical order**.

a) Solve the problem using **Depth-First Search. [5]**

b) Solve the problem using **Breadth-First Search**. **[5]**

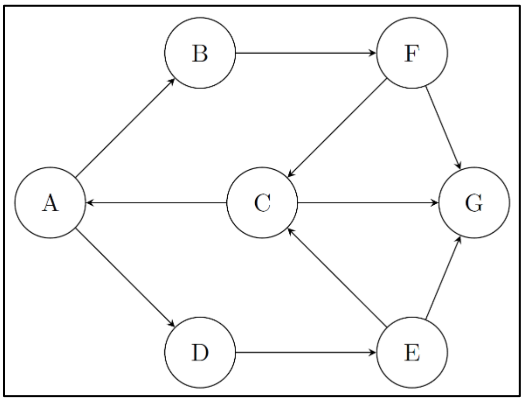

Figure 1: Routes from Gate (A) to Event (G)

CLO 4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.

Q3: Creating a clash-free timetable has always been a challenging task for the academic office. The high influx of students and the limited number of classrooms make this process even more difficult. The schedule must ensure that no two classes are assigned to the same room at the same time and that each class is allocated a room that can accommodate all its students. **[10]**

Mr. Sohaib, an officer in the academic office, attempted to automate this process using technology. He developed a system that manages scheduling while considering these constraints. However, there is still room for improvement, which can be achieved through AI. As an AI expert, you volunteer to help Mr. Sohaib in improving the system. The goal is to create the best possible schedule with no conflicts.

1. **What would be an appropriate objective function for this problem?** Define a function that minimizes scheduling conflicts while ensuring proper classroom assignments. **[3]**
2. **If Mr. Sohaib uses Hill Climbing to find the best classroom assignments, what problems could occur?** Think about how the algorithm might get stuck and why. **[4]**
3. **How could Simulated Annealing help make a better schedule compared to Hill Climbing? [3]**

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